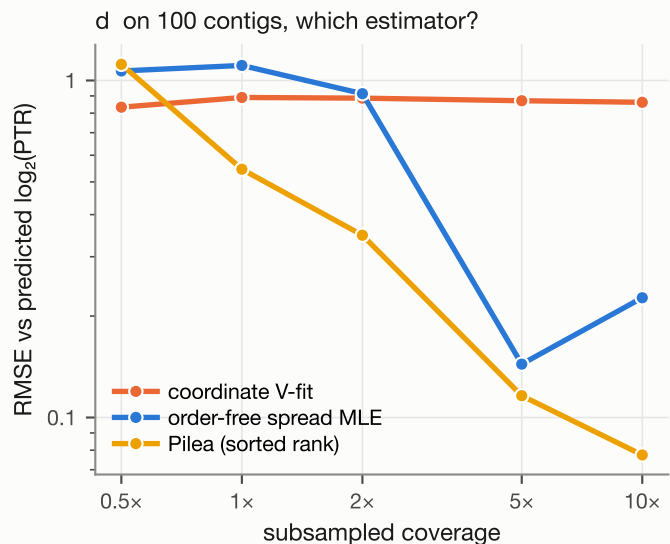
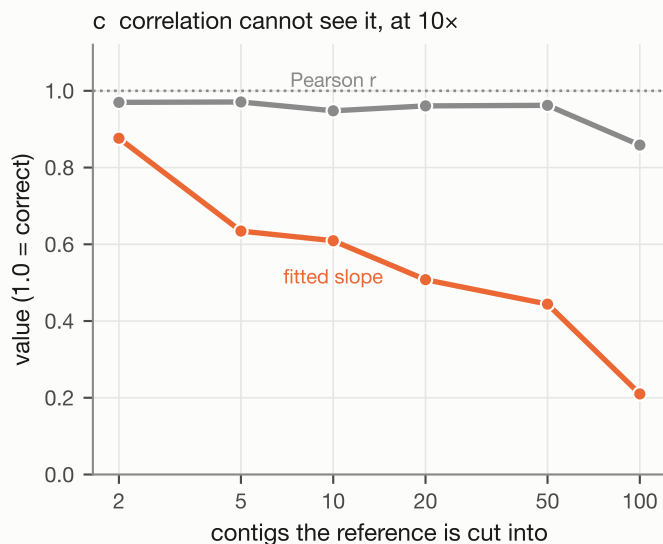
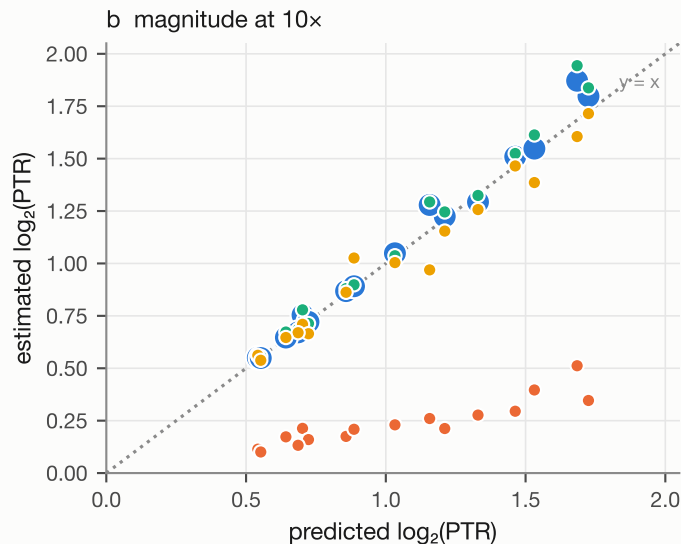
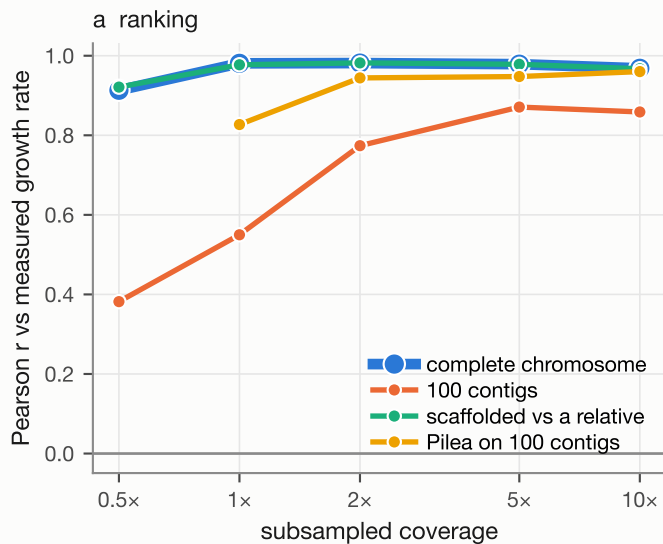




The same reads throughout, and 43,707 of the complete genome's 43,735 anchors survive the cut; only the coordinate changes. Fragmenting into 100 shuffled contigs does not add scatter, it removes the gradient: every estimate collapses toward zero (panel b, slope 0.24), and 100% of those collapsed estimates pass the fusion QC at 5–10x, because a destroyed coordinate makes every enzyme agree there is no gradient. Pileup's rank regression needs no coordinate and is nearly indifferent to the cut, so on unscaffolded contigs it wins outright (r 0.83 against 0.55 at 1x). 'sk2bgrow scaffold' is what puts the coordinate back: it restores the complete-reference curve exactly — the blue halo under the green — even against a different strain. Panel c is why the slope has to be reported beside r : across 2 to 100 contigs the correlation never leaves 0.86–0.97 while the slope falls from 0.88 to 0.21. At 50 contigs — a draft most people would call good, N50 156 kb — r reads 0.96 and every estimate is 44% of truth. Panel d asks whether a coordinate is needed at all. Fitting the distribution of window rates instead of their shape in position — $\log_2$ coverage is uniform across the genome and the width of that uniform is $\log_2(\text{PTR})$ — recovers most of the loss above 5x (RMSE 0.87 to 0.14) with no scaffolding, but carries too little information below it. On the stationary control it reports 0.000 at 1x where Pileup reports 1.153, because the per-window standard error is in the model rather than being read as growth.